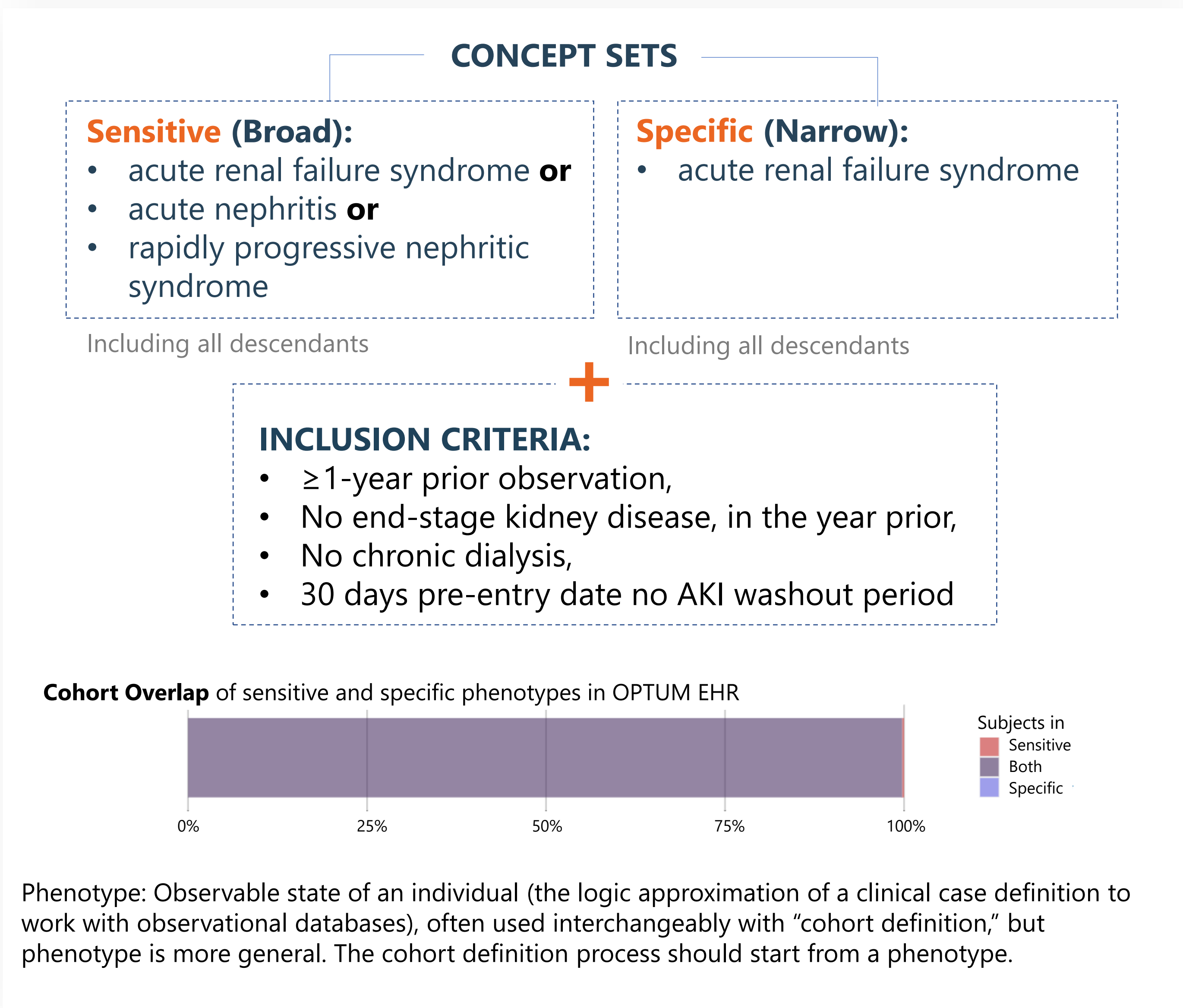


The acute kidney injury phenotype can be used as a cohort definition to generate real-world evidence in data sources with and without lab values

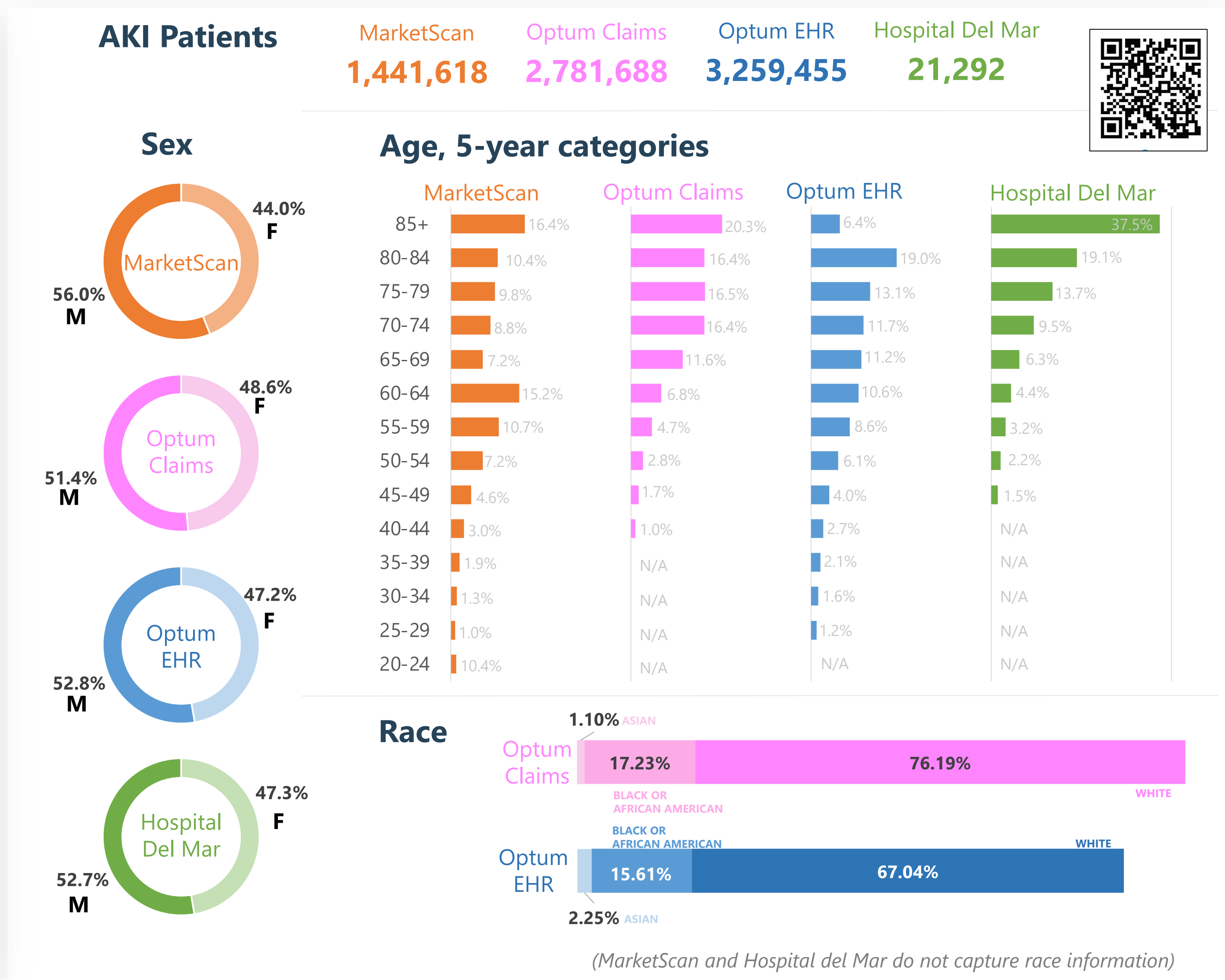
Development and characterization of a phenotype algorithm to identify Acute Kidney Injury in real world data (RWD)

Background: Acute Kidney Injury (AKI) is a common, harmful and potentially treatable disease defined by an abrupt kidney function decrease. Attempts have been made over years to reach consensus on an AKI definition for use in RWD to allow more replicable and reproducible research, but results remain lacking.

Result 1: Proposed phenotypes



Result 2: Demographic characteristics of AKI patients* identified using the proposed specific (narrow) phenotype across databases



Methods

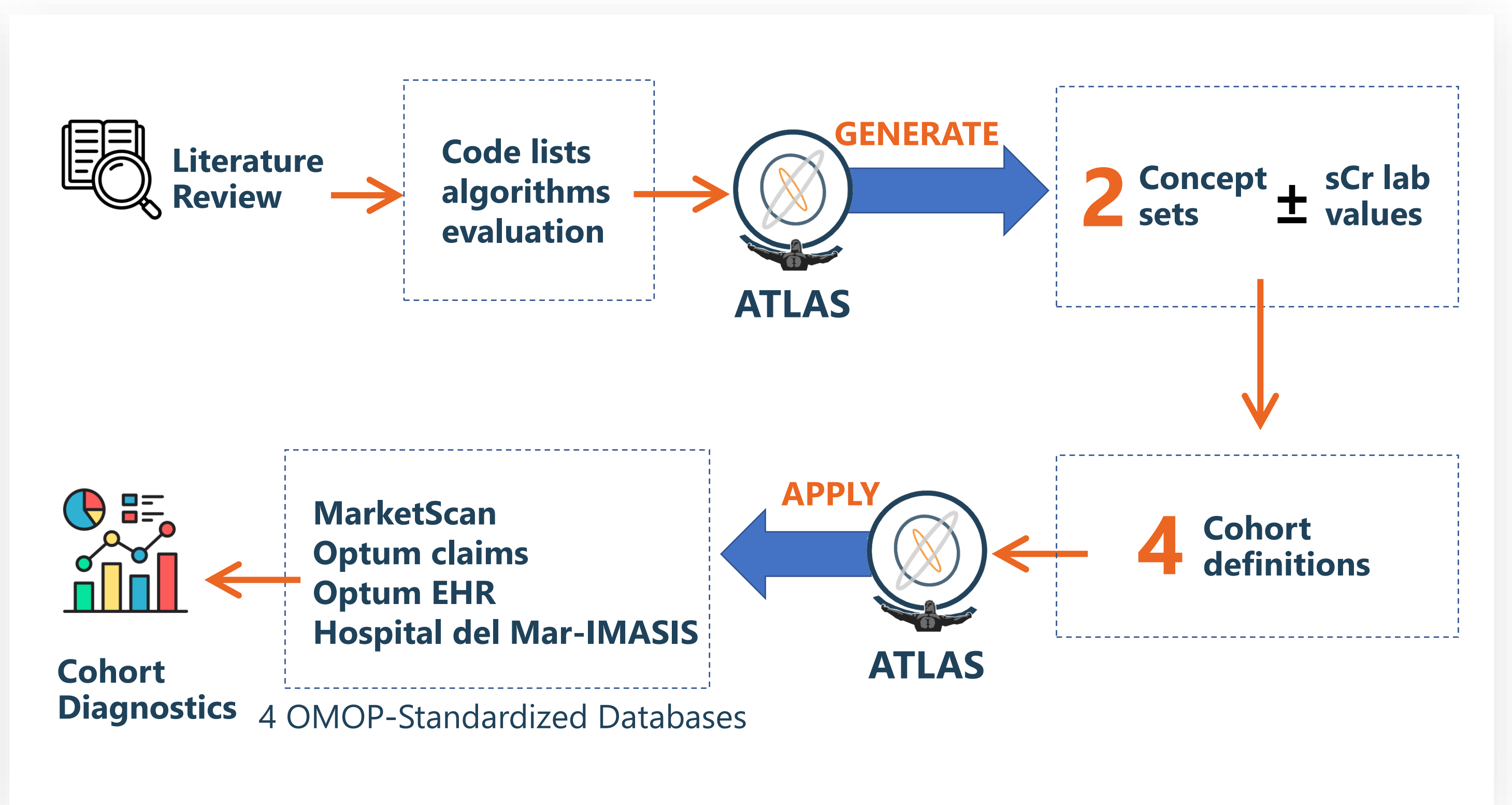
Figure 1. Differential diagnosis of AKI, acute kidney disease (AKD), chronic kidney disease (CKD) & no kidney disease (NKD) [Figure adapted from Kellum** et al. 2012]

	Duration	Functional Criteria	OR	Structural Criteria
AKI	≤7 days	sCr ↑ >50% within 7 days, (sCr ↑ ≥0.3mg/dl within 2 days oliguria for 6h)	OR	–
AKD	<3 months	AKI or GFR <60 ml/min/1.73m ² (GFR ↓ ≥35% over baseline sCr ↑ >50% over baseline)	OR	elevated marker of kidney damage
CKD	>3 months	GFR <60 ml/min/1.73m ²	OR	elevated market of kidney damage
NKD	–	GFR ≥60 ml/min/1.73m ² (stable GFR; stable sCr; no oliguria for ≥6h)	AND	no marker of kidney damage

sCr – serum creatinine; GFR – glomerular filtration rate

** Kellum JA, Lameire N, Aspelin P, Barsoum RS, Burdmann EA, Goldstein SL, et al. Kidney disease: improving global outcomes (KDIGO) acute kidney injury work group. KDIGO clinical practice guideline for acute kidney injury. Kidney International Suppl. 2012;2(1):1–138

Figure 2. Phenotype development approach***



Limitation: Measurement variations relative to a baseline are not supported as an entry event in ATLAS, thus clinical criteria based on serum creatinine (sCr) variation with respect to a baseline cannot be established in ATLAS cohort definitions without flawed workarounds. Sparsity in the data and lack of standardization of lab measurements units did not allow the use of laboratory values in the proposed phenotypes.



Marcela Rivera¹, David Vizcaya², Juan Manuel Ramírez-Anguita³, Azza Shoaibi^{4,5}, Gowtham Rao^{4,5}, Angela Leis³, Miguel Angel Mayer³

¹ Janssen R&D Data Science and Digital Health, Barcelona, Spain; ² Bayer Pharmaceuticals, Sant Joan Despi, Spain; ³ Parc de Salut Mar, Hospital del Mar Medical Research Institute, Barcelona, Spain; ⁴ Janssen Research & Development, LLC, 1125 Trenton-Harbourton Road, Titusville, NJ, 08560, USA; ⁵ Observational Health Data Sciences and Informatics, OHDSI Collaborators, New York, NY, USA